Tashvi Bansal

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Education

Georgia Institute of Technology — Atlanta, GA

Bachelor of Science in Computer Engineering — *Aug. 2026 – May 2029*

- **Relevant Coursework:** Introduction to Object-Oriented Programming, Digital System Design (Georgia Tech); C/C++ Programming I, Data Structures and Algorithms in Java (UCSD Extension)

Experience

Research Intern: LLM Reliability & Formal Reasoning — Feb. 2026 – May 2026

Artificial Intelligence Explainability Accountability (AIEA) Lab | Prof. Leilani Gilpin — *Santa Cruz, CA*

- Co-authored "Know Your Limits: On the Faithfulness of LLMs as Solvers and Autoformalizers in Legal Reasoning"; paper accepted to the AI4Law and AI for Math Workshops at ICML 2026
- Exposed a critical gap between LLM accuracy and logical reliability by benchmarking 3 reasoning approaches across 5 LLMs; identified "scope laundering" in all 5 models, with rates ranging 15.3% (GPT) to 52.5% (Qwen)
- Achieved 19.9% improvement in Claude's legal entailment accuracy using first-order logic reasoning

Research Intern: Neuro-Symbolic AI — June 2025 – October 2025

Artificial Intelligence Explainability Accountability (AIEA) Lab | Prof. Leilani Gilpin — *Santa Cruz, CA*

- Co-authored "Follow My Lead: Logical Fallacy Classification with Knowledge-Augmented LLMs"
- Improved LLM accuracy in classifying logical fallacies by 20.7% by developing a neurosymbolic AI approach using Prolog-based relational graphs and stepwise instructions (i.e. without expensive model training)
- Built and evaluated AWS Bedrock Knowledge Bases for retrieval-augmented reasoning
- Presented the paper at 2025 Advances in Cognitive Systems Conference at the Georgia Institute of Technology and at 2025 Bay Area Machine Learning Symposium at Santa Clara University

Monta Vista Robotics Team — June 2023 – March 2026

Software Engineer | Java, CTRE Phoenix 6, WPILib, SolidWorks, CAN bus — *Cupertino, CA*

- Programmed Java state machine logic using WPILib to control robot intake modes based on real-time sensor feedback; implemented gamepiece intake using limit switches and PID control
- Co-designed a dual-roller intake with sensors for gamepiece detection in SolidWorks, using belts and gears to drive counter-rotating rollers; intakes game pieces in under 6 seconds and competed in 4 matches
- Programmed outreach page for MVRT's official website using HTML, CSS, JavaScript, Bootstrap, and jQuery

Projects

RISC-V CPU | *SystemVerilog, RISC-V, Cadence, Synopsys Design Compiler* — Aug 2026 – Present

- Designed and implemented a RISC-V CPU in SystemVerilog, building RTL for arithmetic, load/store, branch, and shift instructions and integrating instruction/data SRAM
- Verified the CPU using directed and constrained-random testing, assertions, and scoreboarding; detected and characterized all 3 intentionally injected hardware bugs through waveform analysis
- Debugged RTL with Cadence Xcelium and SimVision and analyzed code coverage with Cadence IMC
- Converted the CPU's SystemVerilog RTL into a physical chip layout using Synopsys Design Compiler and Cadence Innovus

GreenOvation iOS App | *Swift, Firebase, CoreML, YOLOv3, Inception* — June 2022 – Apr 2023

- Developed an iOS sustainability app integrating Inceptionv3 and YOLO computer vision models for waste classification and carbon impact analysis, alongside CoreML-based sentiment classification
- Built a Firebase backend for user authentication, Cloud Firestore data storage, and leaderboard tracking; reached 100+ users
- Won Most Creative Hack at PoweringSTEMHacks2 and 3rd place in the Congressional App Challenge (OR-06)

Technical Skills

Languages: C++, Java, Python, JavaScript, HTML, CSS, Swift, SystemVerilog, Prolog
Frameworks & Libraries: Bootstrap, jQuery, SwiftUI, UIKit, CoreML, Firebase, WPILib
Hardware: RTL Design, FPGA Design, RISC-V, CAN Bus, Synthesis, Static Timing Analysis
Tools: AWS Bedrock, Git, GitHub, Z3, Cadence Xcelium, SimVision, IMC, Innovus, Synopsys Design Compiler, AMD Vivado